

Severe Asthma Exacerbation



Section I: Scenario Demographics

Scenario Title:	Severe Asthma Exacerbation		
Date of Development:	01/01/2014		
Target Learning Group:	☐ Juniors (PGY 1 - 2)	☐ Seniors (PGY $\geq$ 3)	☒ All Groups

Section II: Scenario Developers

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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To be exposed to the management of severe asthma requiring third line agents and intubation.
CRM Objectives:	Leads team effectively through crisis Prioritizes tasks and anticipates next steps Communicates clearly and concisely with team regarding orders and plan
Medical Objectives:	Recognize severe respiratory distress and initiate first line treatments while obtaining further history. Administer second and third line treatments for severe asthma Demonstrate a safe approach to the intubation of a severe asthmatic Prevent and/or recognize and appropriately treat air trapping in the ventilated asthmatic patient including premedication and post-intubation ventilation

Case Summary: Brief Summary of Case Progression and Major Events
22 y.o. male is brought by EMS to the emergency department with increasing SOB and chest tightness x12 hours with rapid deterioration over the last hour resulting from a severe asthma exacerbation. He will require multiple pharmaceutical treatments, rapid sequence intubation and proper ventilation.

References
Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). <i>Rosen's emergency medicine: Concepts and clinical practice</i> . St. Louis: Mosby.
Hodder R, Loughheed D, Rowe B, Fitzgerald JM, Kaplan AG, et al. Management of acute asthma in adults in the emergency department: nonventilatory management. <i>CMAJ</i> 2010;282: E55-67. http://www.cmaj.ca/content/182/2/E55.full

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Section IV: Scenario Script

A. Clinical Vignette: To Read Aloud at Beginning of Case			
<i>(vignette delivered by ER RN)</i> A patient has been brought in by EMS and triaged to a Resuscitation Room in the Emergency Department with shortness of breath. He has had an upper respiratory tract infection with cough for 4 days. He's now been having increasing SOB and chest tightness for 12 hours. He may have had a fever yesterday. EMS was called by a housemate who found him struggling to breathe at home. He is no longer responding to ventolin (using 4 puffs q30 min) and has rapidly worsened over the last hour. EMS reported vitals are HR 140, RR 41, O₂Sat 85% on 100% O₂ with face mask.			
B. Scenario Cast & Realism			
Patient:	☐ Computerized Mannequin ☒ Mannequin ☐ Standardized Patient ☐ Hybrid ☐ Task Trainer	Realism: <i>Select most important dimension(s)</i>	☒ Conceptual ☒ Physical ☐ Emotional/Experiential ☐ Other: ☐ N/A
Confederates		Brief Description of Role	
ER RN	Provides history and responds to questions about patient's status. If asked about exam or physical findings: "Severe dyspnea with accessory muscle use, looks tired"		
C. Required Monitors			
☒ EKG Leads/Wires	☒ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
D. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☐ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☒ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☒ Other: Difficult airway cart	
☐ Intraosseous Set-up	☐ LMA	☒ Other: BiPAP machine	
☐ Other: Nebulizer			
E. Moulage			
Mannequin sitting-up, gown on with pants, monitor not attached, IV in place but covered up, spray with sweat.			
F. Approximate Timing			
Set-Up:	3 min	Scenario:	12 min
		Debriefing:	15 min

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Section V: Patient Data and Baseline State

A. Patient Profile and History					
Patient Name: Dale McMurray		Age: 22		Weight: unknown	
Gender: ☒ M ☐ F		Code Status: full			
Chief Complaint: Dyspnea x12 hours					
History of Presenting Illness: (provided by RN at start of simulation) Upper respiratory tract infection (cough) for 4 days, increasing SOB and chest tightness for 12 hours, ?fevers yesterday, EMS was called by a housemate who found him struggling to breathe at home. He is no longer responding to ventolin (using 4 puffs q30 min) and has rapidly worsened over the last hour.					
Past Medical History:	Asthma		Medications:	Ventolin	
	4 previous ER visits			Flovent	
	1 hospital admission (1 year ago)				
Allergies: Penicillin					
Social History: Queen's University Student, from Kitchener originally					
Family History: None contributory					
Review of Systems:	CNS:	Normal			
	HEENT:	Normal			
	CVS:	Normal			
	RESP:	1 word dyspnea			
	GI:	Normal			
	GU:	Normal			
	MSK:	Normal	INT:	Normal	
B. Baseline Simulator State and Physical Exam					
☐ No Monitor Display		☐ Monitor On, no data displayed		☒ Monitor on Standard Display	
HR: 145/min		BP: 140/85mmHg		RR: 36/min	
Rhythm: sinus tach		T: 37.8°C		O ₂ Sat: 87% on 100% FM	
		Glucose: 9.5mmol/L		GCS: 4 5 6 (E V M)	
General Status: respiratory distress					
CNS:	Normal				
HEENT:	Eyes open, PERL, anatomy appears normal, Mallampati 1 with a complete view of the posterior oropharynx, 3 finger mouth opening, normal thyromental distance				
CVS:	Sinus tachycardia				
RESP:	Diffuse quiet wheeze with minimal air entry				
ABDO:	Normal				
GU:	Normal				
MSK:	Normal	INT:	Diaphoretic		

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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: sinus tach HR: 145/min BP: 140/85 RR: 36/min O ₂ SAT: 87% 100% T: 37.8°C	Awake, diaphoretic, wheezing with quiet breath sounds.	<u>Learner Actions</u> - ☐ History & physical - ☐ IV, O ₂ , monitors - ☐ Order blood work & CXR - ☐ IV NS 1L bolus ± Antibiotics - ☐ Temperature check (37.8) - ☐ Glucose check (9.5) - ☐ Start nebulized ventolin & atrovent	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> <u>Triggers</u> <i>For progression to next state</i> - 2 min → 2. Worsening #1 - Intubation → 4. Intubation
2. Worsening #1 HR → 155 BP → 130/70 RR → 42 O ₂ SAT → 89%	Wheezing louder.	<u>Learner Actions</u> - ☐ Continuous nebulized salbutamol 5mg - ☐ Continuous nebulized ipatropium 0.5mg - ☐ IV steroid (eg - methylprednisolone 125mg) - ☐ Magnesium sulfate 2g over 15-20 minutes	<u>Modifiers</u> <u>Triggers</u> - 4 min → 3. Worsening #2 - Intubation → 4. Intubation
3. Worsening #2 BP → 125/65 RR → 35 O ₂ SAT → 85%	Eyes closed, silent chest.	<u>Learner Actions</u> - ☐ Epinephrine 1:1000 IM 0.2-0.5mL up to 3 times - ☐ ± Bipap (bridge to intubation)	<u>Modifiers</u> - If no intubation by 7 minutes → RN to state "I think he's starting to get tired." <u>Triggers</u> - Intubation → 4. Intubation
4. Intubation Unchanged from above.		<u>Learner Actions</u> - ☐ ± Lidocaine 1.5mg/kg iv pre-tx - ☐ Ketamine 1-2mg/kg iv - ☐ Succinylcholine 1-2mg/kg iv - ☐ Intubation - ☐ Dictates ventilator settings (TV 6-8mL/kg, Vent Rate <10bpm, 1:E ratio 1:4, low PEEP 5cmH ₂ O) - ☐ Consult ICU	<u>Modifiers</u> - After intubate → RN asks "what vent settings?" - Post intubation → O ₂ 93%, RR to rate set by learner, HR 120 <u>Triggers</u> - Appropriate ventilation → End Case (ICU arrives) - If RR set >10, wait 90 sec then → 5. Air trapping
5. Air trapping BP → 85/50 O ₂ → 85%	Difficult ventilation. Poor chest movement.	<u>Learner Actions</u> - ☐ Check for symmetrical air entry - ☐ Disconnect ventilator and compress chest - ☐ Adjust ventilator settings	<u>Modifiers</u> - If compress chest → BP 85/50, O ₂ 93% <u>Triggers</u> - Appropriately adjusts ventilator settings → End Case - 12 minutes → End Case

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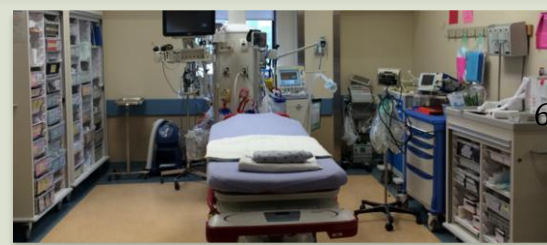


Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results					
VBG	pH: 7.23	PCO ₂ : 59mmHg	PO ₂ : 18mmHg	HCO ₃ : 24mmol	

Images (ECGs, CXRs, etc.)	
CXR: Hyperinflation Image source: http://www.mypacs.net/cases/ACUTE-SEVERE-ASTHMA-ON-31-YO-CXR-3547838.html	

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Section VIII: Debriefing Guide

General Debriefing Plan	
☒ Individual	☐ Group ☐ With Video ☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) How did you determine which drugs were necessary for this patient? 2) At what point did you realize that intubation was necessary? 3) What factors made (or could have made) intubation difficult? Did you assess these prior to attempting intubation? 4) What was your strategy in determining the ventilator parameters? 5) Do you feel that you utilized your other team members effectively?	
Key Moments	
Recognition of critically ill asthmatic	
Decision to intubate	
Management of air trapping	